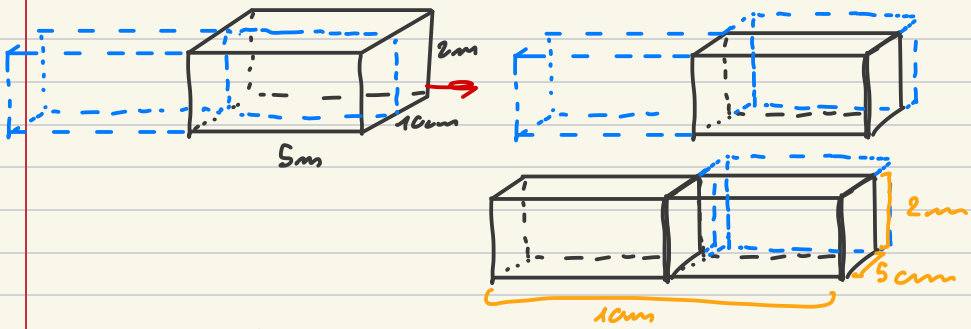
Laminaire :

Méthode Force de Young



PFD {cube} Ecrasement à v = cste

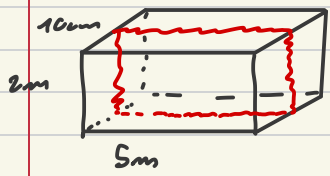
$$0 = \vec{F}_{Hooke} + \vec{F}$$

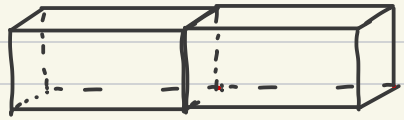
$$F_i = -F_{Hooke}$$
$$= E S \varepsilon_i$$

$$W_i = \int_0^{l_i} \vec{F}_i \cdot d\vec{r} = F_i \cdot l_i = E S_i \varepsilon_i l_i$$

i	S_i	$\frac{\delta L_i}{L_i} = \varepsilon_i$	l_i	W_i
1	$2mm \times 5mm = 10\ mm^2$	$\frac{5mm}{10mm} = \frac{1}{2}$	5cm	5.10^{-10} J
2	$5mm \times 2mm = 10.10^{-2}\ mm^2$	$\frac{5mm}{10mm} = \frac{1}{2}$	5mm	5.10^{-10}

$$W_{tot} = W_1 + W_2 = 100\ GJ$$





on connait

$$E_{rupture} = \text{nbr liaison}\ E_{liaison}$$

$$\text{nbr liaison} = \frac{S_{pavé}}{S_{liaison}} = \frac{S_{pavé}}{l_0^2}$$

} on connait aussi

$$\Rightarrow E_{rupture} = \frac{S_{pavé}\ E_{liaison}}{l_0^2} = \frac{V_{pavé}\ E_{rupture}}{l_0^3}$$

$$= \frac{2 \times 5 \times 10^{-18}}{(10^{-10})^2}$$

$$= \frac{10^{-18}}{10^{-20}}$$

$$= 100\ J$$

NUL

$$= \frac{10^{-1}}{10^{-30}}\ 1.10^{-18}$$

$$= 10^{11}$$

$$= 100\ GJ$$

mieux mais

je sais pas justifier